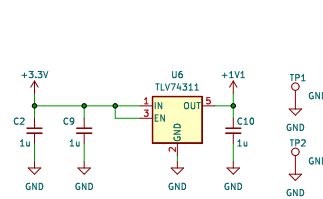
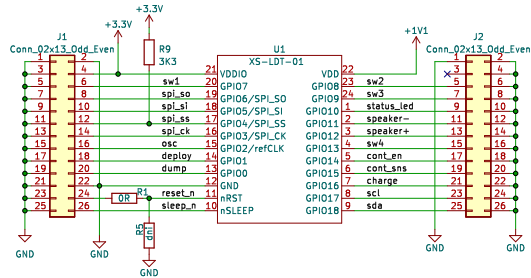
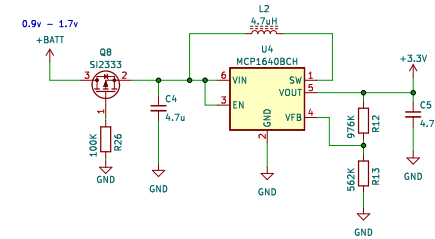


# Launch Deployment Timer

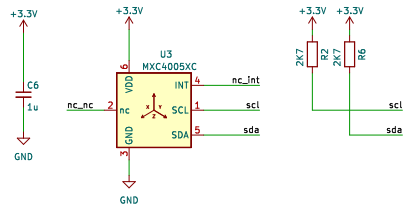


### 3.3v Power

The diagram illustrates a 3.3V power regulation circuit. It begins with a +BATT input, which is connected to a 100K resistor (R26) and the non-inverting input (pin 3) of an op-amp (Q8, Si2333). The op-amp's output (pin 2) is connected to the VIN pin (pin 6) of the MCP1640BCH DC-DC converter (U4). The converter's EN pin (pin 3) is connected to ground. The SW pin (pin 1) is connected to the SW pin (pin 5) of the converter. The VOUT pin (pin 4) is connected to the VFB pin (pin 4) of the converter. The output of the converter is connected to a 978K resistor (R12) and a 562K resistor (R13), which are connected to ground. The output voltage is +3.3V, and a 4.7uF capacitor (C5) is connected to ground.



# Accelerometer

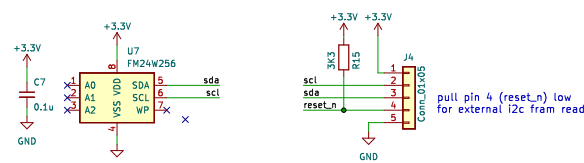


# Flight Recorder Opt.

The left diagram illustrates the internal circuitry of the Flight Recorder Opt. module. It features a 3.3V power supply connected to pin 8 of the U7 FM24W256 chip. A 0.1uF capacitor (C7) is connected between the 3.3V supply and ground. The chip's pins are labeled: A0, A1, A2 (address lines), VSS, VDD (power), SDA, SCL (I2C lines), and WP (write protect). The SDA and SCL lines are connected to the external SDA and SCL pins of the J4 connector. The WP pin is connected to ground.

The right diagram shows the external connections to the J4 connector. A 3k3 pull-up resistor is connected between the 3.3V supply and the reset\_n pin (pin 4). The 3.3V supply is also connected to pins 1 and 2. The SDA and SCL lines are connected to pins 3 and 5, respectively. The Comm\_Diag05 pin (pin 6) is connected to ground.

pull pin 4 (reset\_n) low  
for external I2c fram read



# Front Panel

The front panel circuit includes a speaker and a status LED. The speaker is connected to the `speaker+` and `speaker-` pins of the microcontroller. The status LED is connected to the `status_led` pin through a diode (D3) and a resistor (R4, 1K).

**Speaker Circuit:**

- `speaker+` (green line) connects to pin 1 of the speaker.
- `speaker-` (red line) connects to pin 2 of the speaker.
- The speaker is labeled `LS1 Piezo`.

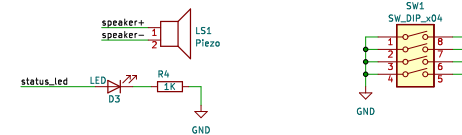
**Status LED Circuit:**

- `status_led` (green line) connects to the anode of diode `D3`.
- The cathode of `D3` connects to the LED pin of the microcontroller.
- The LED pin also connects to resistor `R4` (1K).
- Resistor `R4` connects to `GND`.

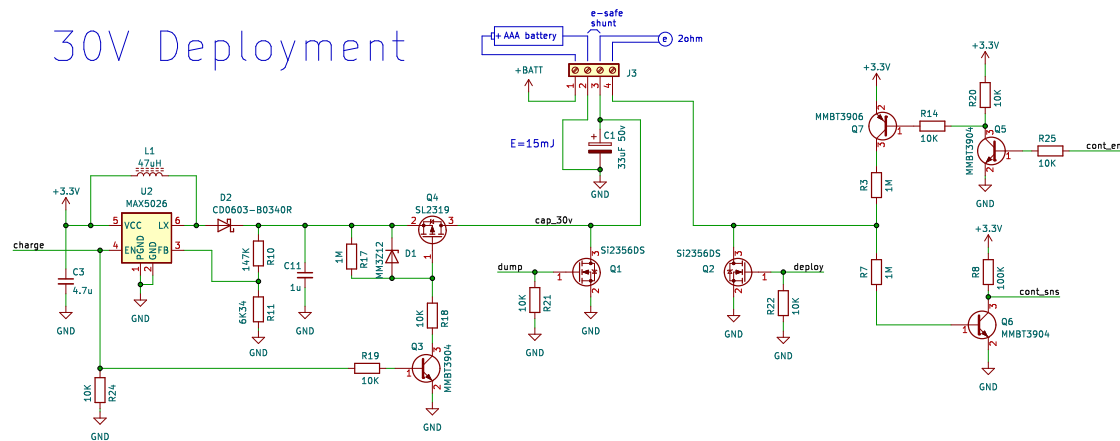
The front panel also includes four switches connected to the `SW_DIP_x04` pins of the microcontroller. The switches are labeled `sw1`, `sw2`, `sw3`, and `sw4`.

**Switches Circuit:**

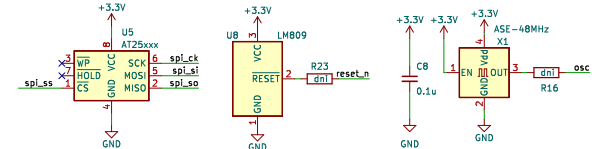
- The switches are connected to pins 1, 2, 3, and 4 of the `SW_DIP_x04` component.
- The other side of the switches connects to pins 8, 7, 6, and 5 of the microcontroller.
- All switch pins are connected to `GND`.



# 30V Deployment



# Cut-away parts



A graphic titled "Launch Deployment Timer" in large blue font. Below the title, it lists "Eric Pearson, XYZZY Semiconductor" in red. It includes a "Sheet: /" label, a "File: dev\_board.kicad\_sch" label, and a "Title: X5-LD7-DB Development Board" label. At the bottom, it shows "Size: B" and "Date: 2026-09-22" on the left, and "KiCad E.D.A. kicad 7.0.2" and "Rev: V1 Id: 1/1" on the right.

Id: 1/1